Lecture 19: Social media and democracy

Matthew J. Salganik

Sociology 204: Social Networks
Princeton University

April 7, 2025

Logistics:

- How's the self-experiment going?

Social media:

- Lecture 17: Social media and individuals
- Lecture 18: Social media and society
- Lecture 19: Social media and democracy
- Lecture 20: Fixing social media

Review:

- Social media seems to amplify moral outrage and false rumors

Review:

- Social media seems to amplify moral outrage and false rumors
- This happens because of the choices of people and the choices of social media architects

Social media:

- Lecture 17: Social media and individuals
- Lecture 18: Social media and society
- Lecture 19: Social media and democracy
- Lecture 20: Fixing social media

1. Guess et al. (2023). How do social media feed algorithms affect attitudes and behavior in an election campaign? *Science*.
2. González-Bailón et al. (2024) The Diffusion and Reach of (Mis)Information on Facebook During the U.S. 2020 Election. *Sociological Science*.

Background

2016 United States presidential election

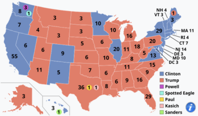
← 2012 | **November 8, 2016** | 2020 →

538 members of the Electoral College
270 electoral votes needed to win

Opinion polls

Turnout 60.1%[1] (▲ 1.5 pp)

Nominee	**Donald Trump**	Hillary Clinton
Party	Republican	Democratic
Home state	New York	New York
Running mate	**Mike Pence**	Tim Kaine
Electoral vote	**304**[a]	227[a]
States carried	**30 + ME-02**	20 + DC
Popular vote	62,984,828[2]	**65,853,514**[2]
Percentage	46.1%	**48.2%**



Presidential election results map. Red denotes states won by Trump/Pence and blue denotes those won by Clinton/Kaine. Numbers indicate electoral votes cast by each state and the District of Columbia. On election night, Trump won 306 electors and Clinton 232. However, because of seven faithless electors (five Democratic and two Republican), Trump received 304 votes and Clinton 227.

https://en.wikipedia.org/wiki/2016_United_States_presidential_election

There were many questions about the role of social media, especially Facebook, in this election. This is hard to study for many reasons. Straud et al. propose two:

- companies have legal and commercial reasons for wanting to keep their data and algorithms secret

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- conflicts of interest

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- conflicts of interest
- serious logistical difficulties for outside research to understand and use the Facebook systems

So a new partnership is formed . . . .

> With these goals in mind, Facebook today announced a new project that is bringing together a team of approximately two-dozen Facebook researchers and outside scholars to study Facebook's and Instagram's impact on four key outcomes that have dominated public and academic attention: political participation, political polarization, knowledge and misperceptions, and trust in US democratic institutions.

Note the scope of this partnership.

This was a serious attempt with a careful design . . .

```
- A rapporteur with access to all researchers who will document the
research process

- Pre-registration of study plans before data collection that will be
made public simultaneously with the publication of results

- Explicit informed consent of study participants whose individual-
level data will be collected and analyzed

- Listing all researchers who have contributed to the project as co-
authors on papers, with clear delineation of their contribution to
that paper

- An attempt to publish all pre-registered study designs in peer
reviewed journals; those that cannot be published will be posted in
public scholarly archives

- Processes to provide journals publishing the papers with the
ability to replicate the analyses in the papers

- Publication of all papers accepted by peer reviewed journals in an
Open Access format, which means they will be freely available to the
public

- As much of the data generated from the project as is possible --
given privacy limitations -- made available to scholars for further
study in the future.
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https://medium.com/@2020_election_research_project/a-proposal-for-understanding-social-medias-impact-on-elections-4ca5b7aae10

2020 United States presidential election

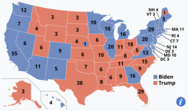
← 2016 **November 3, 2020**[a] 2024 →

538 members of the Electoral College
270 electoral votes needed to win

Opinion polls

Turnout 66.6% (▲ 6.5 pp)[b]

Nominee	**Joe Biden**	Donald Trump
Party	Democratic	Republican
Home state	Delaware	Florida[c]
Running mate	**Kamala Harris**	Mike Pence
Electoral vote	**306**	232
States carried	**25 + DC + NE-02**	25 + ME-02
Popular vote	**81,283,501**[1]	74,223,975[1]
Percentage	**51.3%**	46.8%



Presidential election results map. Blue denotes states won by Biden/Harris and red denotes those won by Trump/Pence. Numbers indicate electoral votes cast by each state and the District of Columbia.

https://en.wikipedia.org/wiki/2020_United_States_presidential_election

How do social media feed algorithms affect attitudes and behavior in an election campaign?

Andrew M. Guess[1*], Neil Malhotra[2], Jennifer Pan[3], Pablo Barberá[4], Hunt Allcott[5], Taylor Brown[4], Adriana Crespo-Tenorio[4], Drew Dimmery[4,6], Deen Freelon[7], Matthew Gentzkow[8], Sandra González-Bailón[9], Edward Kennedy[10], Young Mie Kim[11], David Lazer[12], Devra Moehler[4], Brendan Nyhan[13], Carlos Velasco Rivera[4], Jaime Settle[14], Daniel Robert Thomas[4], Emily Thorson[15], Rebekah Tromble[16], Arjun Wilkins[4], Magdalena Wojcieszak[17,18], Beixian Xiong[4], Chad Kiewiet de Jonge[4], Annie Franco[4], Winter Mason[4], Natalie Jomini Stroud[19], Joshua A. Tucker[20]

Published 28 July 2023

Algorithmic feed vs Chronological feed

Shedding More Light on How Instagram Works

By Adam Mosseri · June 8, 2021

https://about.instagram.com/blog/announcements/shedding-more-light-on-how-instagram-works

What is “the algorithm”?

One of the main misconceptions we want to clear up is the existence of “The Algorithm.” Instagram doesn’t have one algorithm that oversees what people do and don’t see on the app. We use a variety of algorithms, classifiers, and processes, each with its own purpose. We want to make the most of your time, and we believe that using technology to personalize your experience is the best way to do that.

When we first launched in 2010, Instagram was a single stream of photos in chronological order. But as more people joined and more was shared, it became impossible for most people to see everything, let alone all the posts they cared about. By 2016, people were missing 70% of all their posts in Feed, including almost half of posts from their close connections. So we developed and introduced a Feed that ranked posts based on what you care about most.

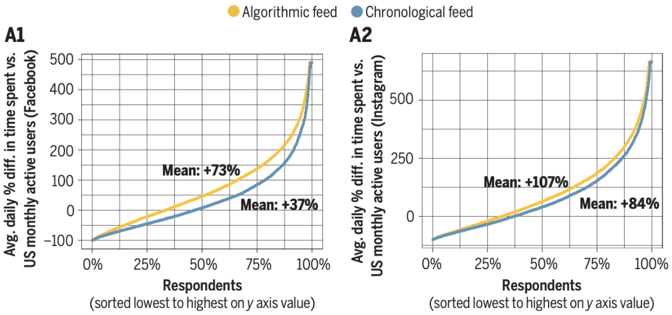
Each part of the app – Feed, Explore, Reels – uses its own algorithm tailored to how people use it. People tend to look for their closest friends in Stories, but they want to discover something entirely new in Explore. We rank things differently in different parts of the app, based on how people use them.

https://about.instagram.com/blog/announcements/shedding-more-light-on-how-instagram-works

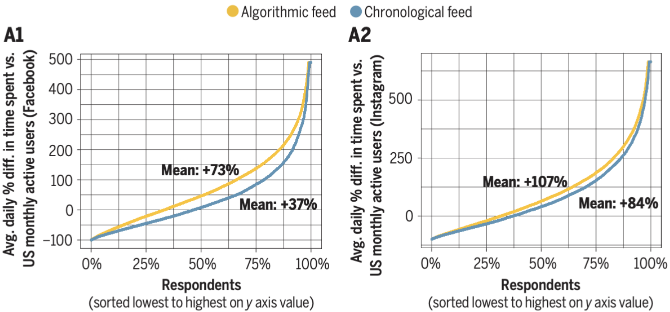
How does the Instagram ranking algorithm work for feed?

How we rank Feed and Stories

- **Information about the post.** These are signals both about how popular a post is – think how many people have liked it – and more mundane information about the content itself, like when it was posted, how long it is if it's a video, and what location, if any, was attached to it.
- **Information about the person who posted.** This helps us get a sense for how interesting the person might be to you, and includes signals like how many times people have interacted with that person in the past few weeks.
- **Your activity.** This helps us understand what you might be interested in and includes signals such as how many posts you've liked.
- **Your history of interacting with someone.** This gives us a sense of how interested you are generally in seeing posts from a particular person. An example is whether or not you comment on each other's posts.



- "We found that users in the Chronological Feed group spent dramatically less time on Facebook and Instagram."



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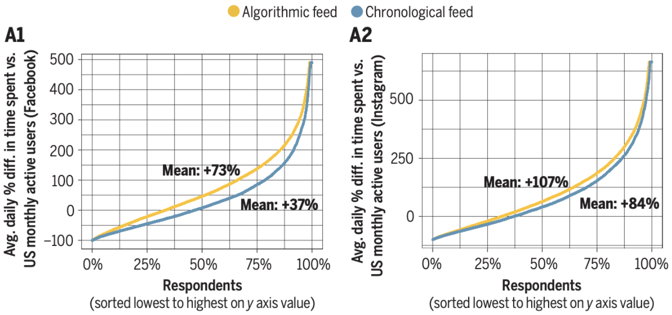
- FB: 73% more than average to 37% more than average (eg 173 minutes per day to 137 minutes per day)



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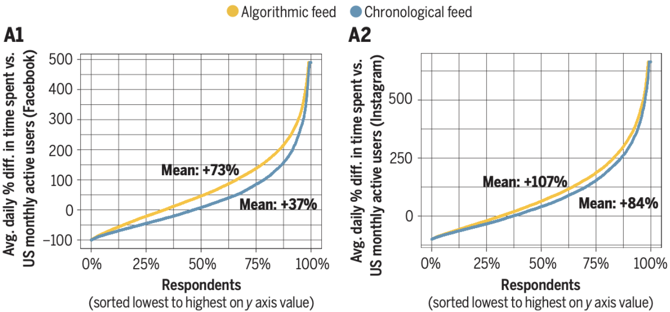

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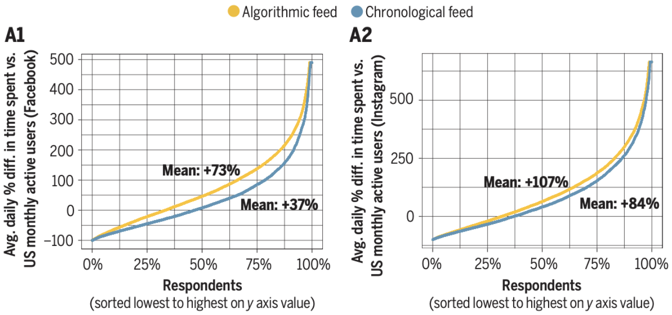
- Substitution: On mobile, Insta users spend 36% more time on TikTok and 20% more time on YouTube. (what does this mean for companies thinking about changes?)



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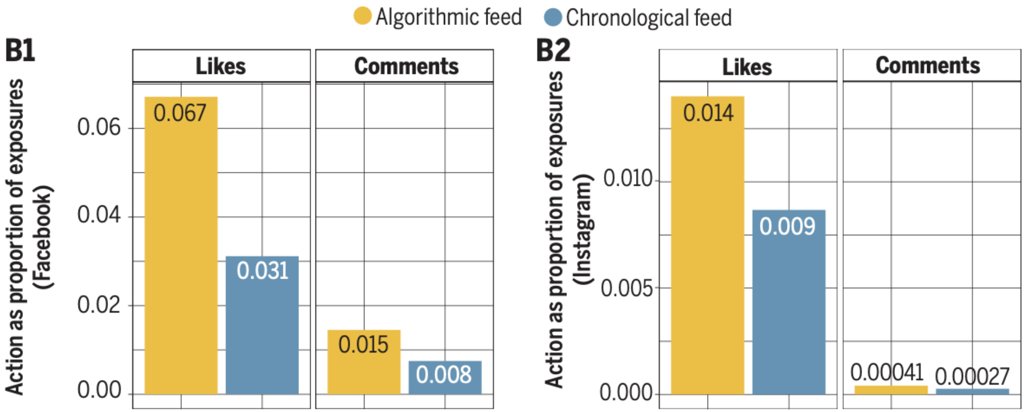
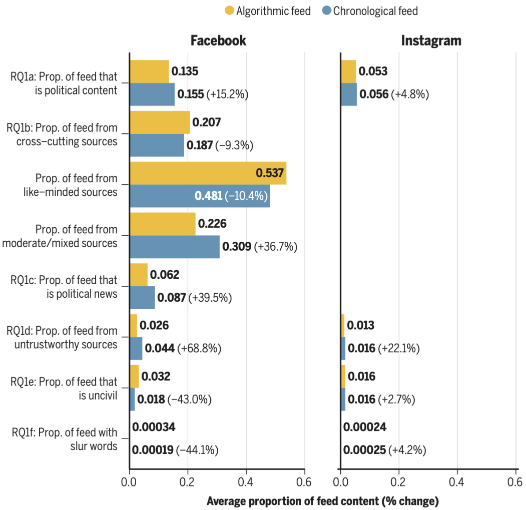
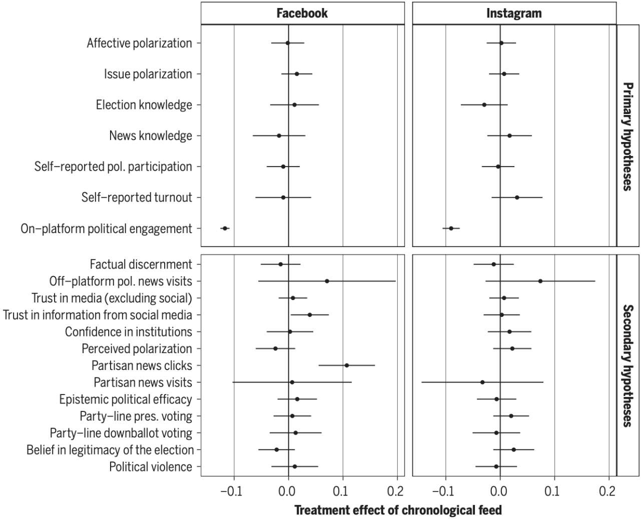


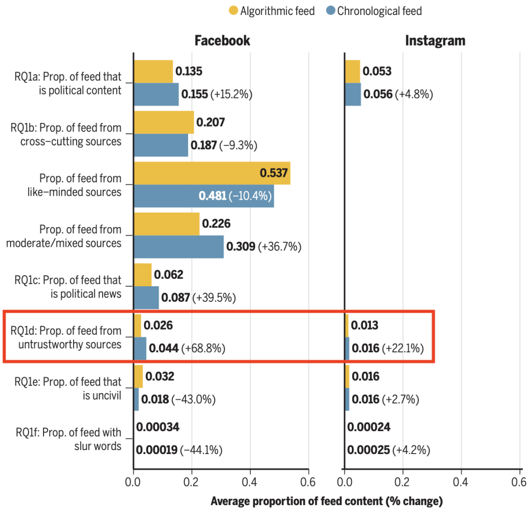
The algorithmic feed is good at finding content that users will like and comment.



Content users see changes



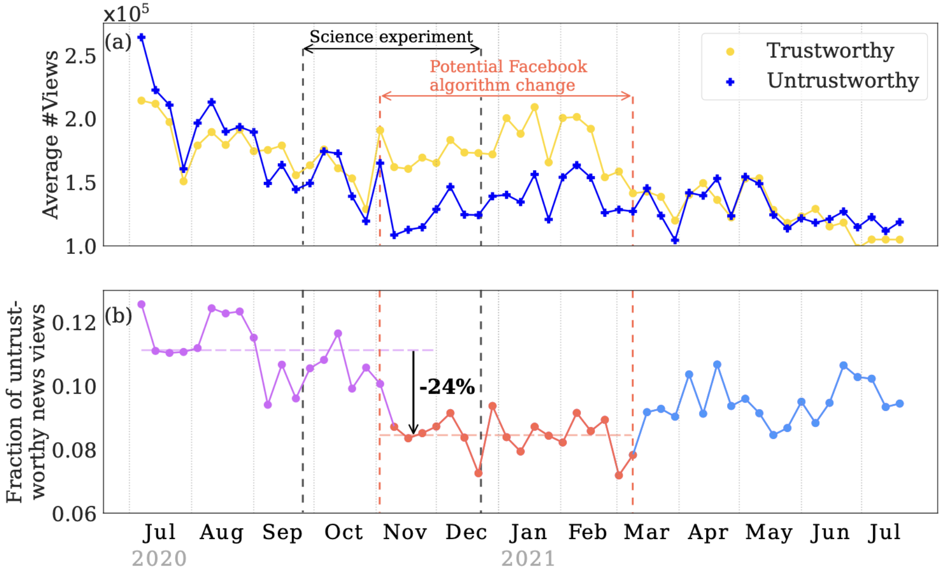
“few discernible changes in political attitudes, knowledge, and offline behaviors”



Content users see changes

Social media algorithms can curb misinformation, but do they?

Chhandak Bagchi[1], Filippo Menczer[2], Jennifer Lundquist[1], Monideepa Tarafdar[1], Anthony Paik[1], Przemyslaw A. Grabowicz[1,3]*



x10^5
(a)
Science experiment
Potential Facebook algorithm change
Trustworthy
Untrustworthy
Average #Views
2.5
2.0
1.5
1.0
(b)
-24%
Fraction of untrust-worthy news views
0.12
0.10
0.08
0.06
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
2020
2021

SEP. 26, 2024

Response to “Social media algorithms can curb misinformation, but do they?”

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In their letter, Bagchi et al. (2024) comment on conclusions that can be reached from a field experiment conducted by Guess et al. (2023) assessing the effect of moving Facebook users from the default algorithmic feed (control condition) to a reverse chronological feed (treatment condition). Bagchi et al. (2024) argue that one must exercise care in interpreting and generalizing the results because Facebook implemented "break-glass measures" to combat misinformation during and in the aftermath of the 2020 U.S. presidential campaign. Specifically, they write: "the fact that the control condition changed during their experiment affects the validity and conclusion of their study."

We first clarify that although the introduction of "break-glass measures" may have impacted the experiment's *external validity* (i.e., whether the findings could be generalized to other contexts), it did not affect its *internal validity* (i.e., whether we were able to reach accurate causal conclusions from the data that were collected). As the purpose of Guess et al. (2023) was to estimate the cumulative causal effect of moving people to a reverse-chronological feed from the Facebook algorithm as it existed over the course of the entire study period (before and after the 2020 U.S. presidential election), the experiment is internally valid. As is common in field experiments, the conditions in the control group may change over the course of the treatment period, but this does not affect the causal inferences that can be drawn regarding the cumulative effect.

"Far better an approximate answer to the *right* question, which is often vague, than an exact answer to the *wrong* question which can always be made precise."
- Johh Tukey (1962), emphasis in original

An additional set of limitations concerns the nature of our design: Our research design provides estimates of direct effects on individuals and as such cannot speak to whether social media shapes societal incentives or influences the behavior of other users. For example, if ranking algorithms affect the demand for certain kinds of content, they could influence the decisions of content producers (such as news organizations), civic organizations, and political campaigns, which would feed back into the inputs of the algorithmic system itself. Even more simply, the Chronological Feed experiences that we studied were composed in part of posts shared by other users not in our sample whose behavior was shaped by the standard algorithmic rankings. Our design thus cannot speak to "general equilibrium" effects. If our randomized intervention (Chronological Feed) were to be scaled up to the population of all users, its impact may differ because algorithmic feedback within networks may generate complex-system dynamics. Future research can examine whether some of the strongest effects uncovered in this study—such as the decreased time spent on platform—mediate the relationship between the Chronological Feed and political attitudes and behaviors.

sociological science

The Diffusion and Reach of (Mis)Information on Facebook During the U.S. 2020 Election

Sandra González-Bailón,[a] David Lazer,[b] Pablo Barberá,[c] William Godel,[c] Hunt Allcott,[d] Taylor Brown,[c] Adriana Crespo-Tenorio,[c] Deen Freelon,[a] Matthew Gentzkow,[d] Andrew M. Guess,[e] Shanto Iyengar,[d] Young Mie Kim,[f] Neil Malhotra,[d] Devra Moehler,[c] Brendan Nyhan,[g] Jennifer Pan,[d] Carlos Velasco Rivera,[c] Jaime Settle,[h] Emily Thorson,[i] Rebekah Tromble,[j] Arjun Wilkins,[c] Magdalena Wojcieszak,[k,l] Chad Kiewiet de Jonge,[c] Annie Franco,[c] Winter Mason,[c] Natalie Jomini Stroud,[m] Joshua A. Tucker[n]

sociological science

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The spread of true and false news online

Soroush Vosoughi,[1] Deb Roy,[1] Sinan Aral[2]*

Affordances: roughly, what is made easy and what is made hard

Affordances: roughly, what is made easy and what is made hard

Our observation window includes periods of high-intensity interventions designed to moderate content on the platform above and beyond what the Feed algorithm is designed to do [S30]. The leading academic authors reconstructed the timeline of extraordinary platform interventions shown in Figure S11 (including the "break the glass" measures discussed in the main text), using three main sources available in the public record: (1) The memo that circulated among the January 6 House Committee members, leaked to the Washington Post [S31]; (2) BuzzFeed's reporting on the Stop the Steal campaign[S32]; and (3) Jeff Horwitz's reporting in his book *Broken Code. Inside Facebook and the Fight to Expose Its Harmful Secrets*[S33]. The academic team requested, and did not receive from Meta, more details about these platform interventions. According to Meta, they did not provide this information "due to adversarial risks and concerns about the interventions being less effective in the future if their details become public".

Jun, 2020 "An internal Facebook intelligence report warned of growing danger from QAnon activity on Facebook" (J6 Social Media Report, p. 31).

Aug 19, 2020 "Facebook removed thousands of groups, pages, accounts, and ads tied to QAnon and various militia groups and acted to reduce the reach and distribution of remaining accounts and hashtags on the platform" (J6 Social Media Report, p. 32).

Sep 22, 2020 "In September, Nick Clegg had told USA Today that the company had prepared contingency plans but would not be discussing them" (Broken Code, p. 215).

- USA Today article: https://www.usatoday.com/story/news/politics/elections/2020/09/22/election-2020-facebook-has-break-glass-measures-if-violence-erupts/5866803002/
- "Facebook' s civic integrity team worked diligently to address competing complex risk areas in advance of the election by developing 63 "break the glass" measures designed to slow the flow of viral, potentially harmful content." (J6 Social Media Report, p. 29).

Sep 30, 2020 By September, the company had stopped "actively boosting non-recommended groups in News Feed" and "established some limits on 'bulk invites'. It also 'imposed a three-week waiting period before newly created groups on any subject were eligible to be recommended to Facebook users" (Broken Code, pp. 209-210).

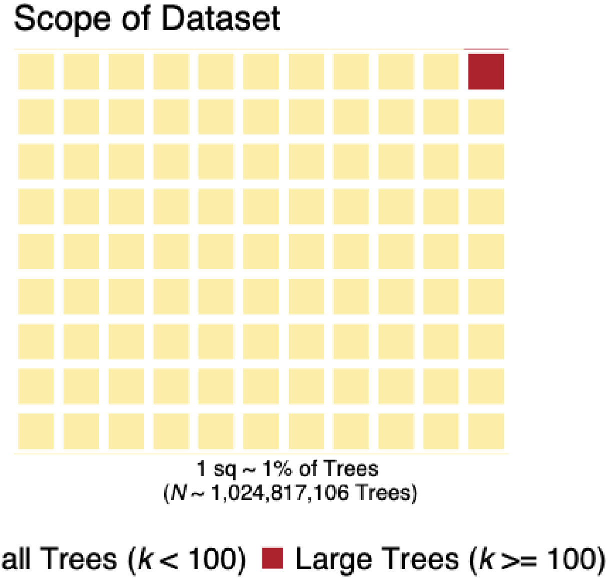
- The 'Groups Task Force' had made a contribution to Civic Integrity's arsenal of Break the Glass interventions: "If all hell broke loose around Election Day, the company could throw a switch and force the administrators of groups with a history of breaking Facebook's rules to manually approve all member posts. The change would both slow down the groups overall and force the admins to be responsible for the content they approved" (Broken Code, p. 210).
- During this month, Facebook announced that it would ban political ads before and after the election (Broken Code, p. 211).
- During this month, Facebook labeled speech by political figures, directing users toward outside sources of factual information about voting; it then declared the election over when ballots had been counted (Broken Code, p. 211).
- During this month, Facebook accepted changing the algorithm so that it would "stop treating anger emojis as grounds to amplify a post" (Broken Code, p. 211).

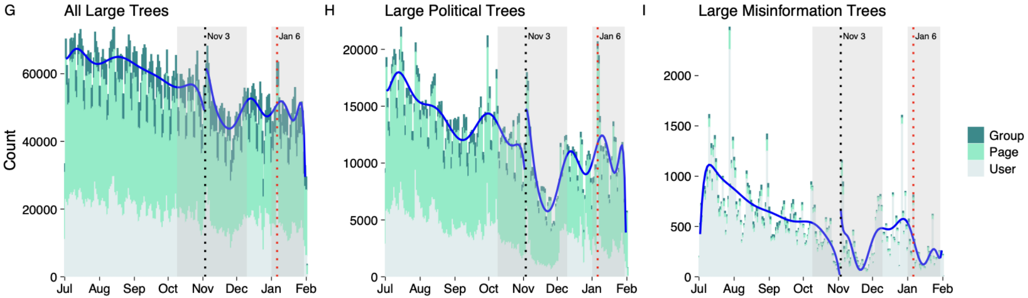
Oct 4, 2020 Break the glass (BtG) measure to remove all Groups created in the last 21 days from Recommendations in order to offset the low recall and detection of Groups potentially associated with violence and other harms is launched and never deprecated (J6 Social Media Report, p. 36).

Oct 9, 2020 BtG measure called "virality circuit breaker" is deployed to slow the distribution of URLs linking to unknown external domains that may contain misinformation (J6 Social Media Report, p. 35 and p. 57).

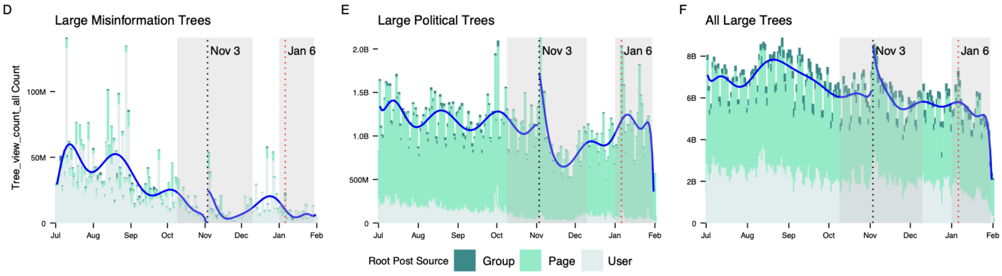
Figure S11: **Timeline of Interventions.** This timeline was compiled by the leading academic authors using publicly available information (1 of 8).

A


Scope of Dataset
1 sq ~ 1% of Trees
(N ~ 1,024,817,106 Trees)
Small Trees (k < 100)
Large Trees (k >= 100)



Misinformation defined by third-party fact checker program (which has now been ended)



Very unusual to have data on views

Both of these papers conclude that more research is needed, but that research is unlikely to happen without structural changes

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Summary:

- Users in Chronological Feed had large changes in online behavior but minimal changes in other behavior and attitudes (relative to Algorithmic feed)
- The Algorithmic feed is not static, and it could be changed
- Misinformation (as determined by 3rd party fact checkers) spread more slowly than other content on FB around around the 2020 US Presidential Election
- “Break the glass” measures by Facebook appear to have had some impact, but it is hard to know

What do you think is the most important unanswered question that could be addressed with cooperation between social media companies and outside researchers?

Social media:

- Lecture 17: Social media and individuals
- Lecture 18: Social media and society
- Lecture 19: Social media and democracy
- Lecture 20: Fixing social media